

Kishore S

CH.SC.U4CSE24222

Week – 4

Design and Analysis of Algorithm(23CSE211)

Merge and Quick Sort

1. Write a c program for Merge Sort

Code:

```
#include <stdio.h>

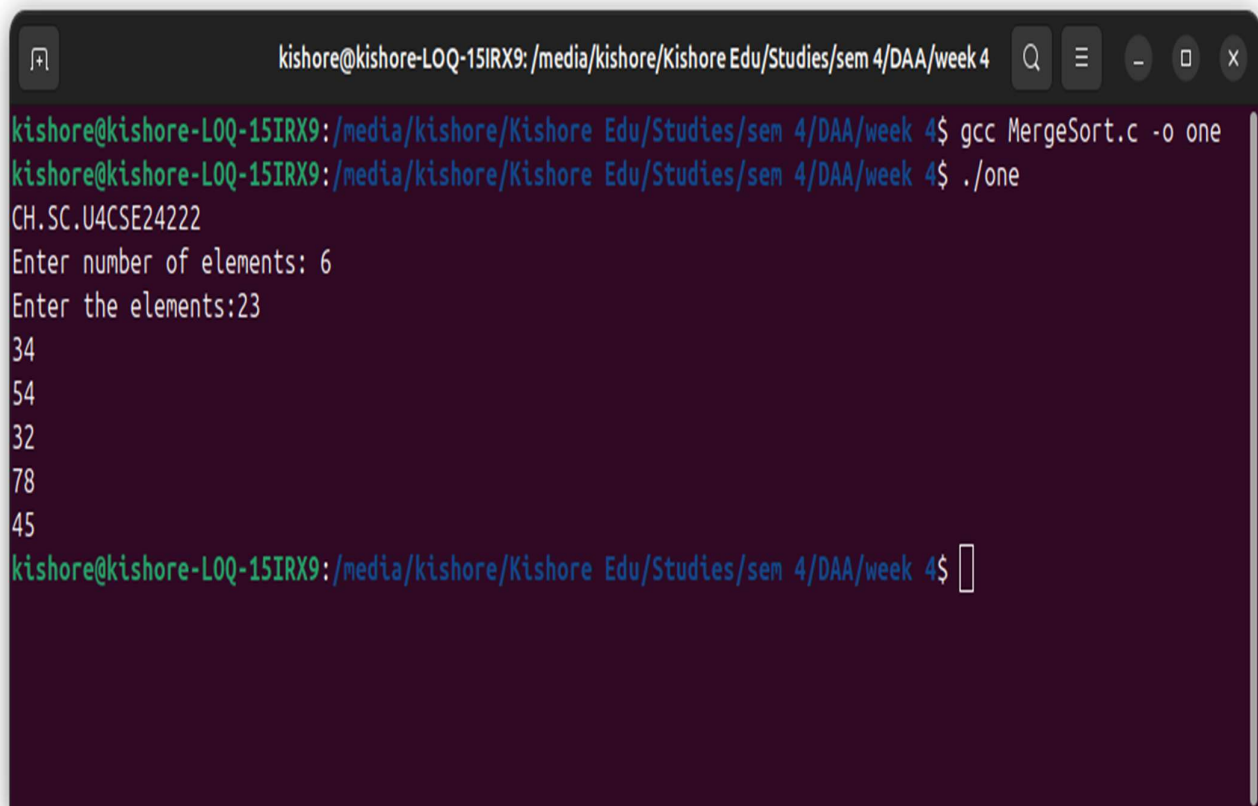
void merge(int arr[], int l, int m, int r) {
    int n1 = m - l + 1;
    int n2 = r - m;
    int L[n1], R[n2];
    for (int i = 0; i < n1; i++) L[i] = arr[l + i];
    for (int j = 0; j < n2; j++) R[j] = arr[m + 1 + j];
    int i = 0, j = 0, k = l;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) arr[k++] = L[i++];
        else arr[k++] = R[j++];
    }
    while (i < n1) arr[k++] = L[i++];
    while (j < n2) arr[k++] = R[j++];
}

void mergeSort(int arr[], int l, int r) {
    if (l < r) {
        int m = l + (r - l) / 2;
        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}

int main() {
    int n;
    printf("CH.SC.U4CSE24222\n");
    printf("Enter number of elements: ");
    scanf("%d", &n);
```

```
int arr[n];  
printf("Enter the elements:");  
for (int i = 0; i < n; i++) scanf("%d", &arr[i]);  
mergeSort(arr, 0, n - 1);  
printf("Sorted array: ");  
for (int i = 0; i < n; i++) printf("%d ", arr[i]);  
printf("\n");  
return 0;  
}
```

OUTPUT:



```
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$ gcc MergeSort.c -o one  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$ ./one  
CH.SC.U4CSE24222  
Enter number of elements: 6  
Enter the elements:23  
34  
54  
32  
78  
45  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$
```

2. Write a c program for Quick Sort

Code:

```
#include <stdio.h>

void swap(int* a, int* b) {
    int t = *a;
    *a = *b;
    *b = t;
}

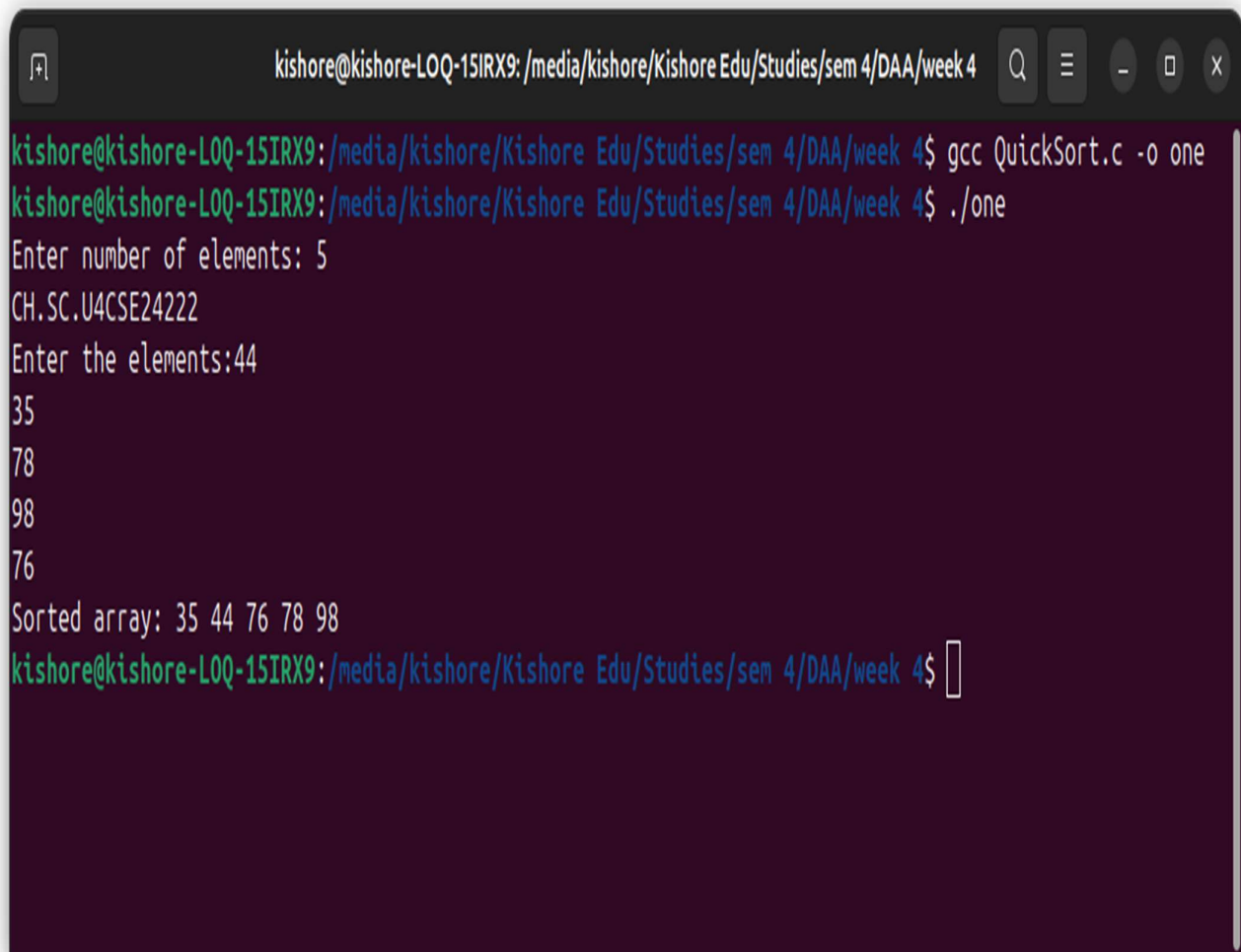
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = (low - 1);
    for (int j = low; j <= high - 1; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

int main() {
    int n;
    printf("Enter number of elements: ");
```

```
scanf("%d", &n);  
printf("CH.SC.U4CSE24222\n");  
int arr[n];  
printf("Enter the elements:");  
for (int i = 0; i < n; i++) scanf("%d", &arr[i]);  
quickSort(arr, 0, n - 1);  
printf("Sorted array: ");  
for (int i = 0; i < n; i++) printf("%d ", arr[i]);  
printf("\n");  
return 0;  
}
```

OUTPUT:



A terminal window with a dark background and light-colored text. The window title bar shows the path `kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4`. The terminal content shows the compilation and execution of a C program. The user enters `5` for the number of elements, followed by `CH.SC.U4CSE24222` as a string. Then, the user enters the elements `35`, `78`, `98`, and `76`. The program outputs the sorted array: `Sorted array: 35 44 76 78 98`. The prompt `kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$` is visible at the bottom.

```
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$ gcc QuickSort.c -o one  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$ ./one  
Enter number of elements: 5  
CH.SC.U4CSE24222  
Enter the elements:44  
35  
78  
98  
76  
Sorted array: 35 44 76 78 98  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 4$
```